

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0232

South Woodrow 100 kV
March 03, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0232. The Interconnection Customer proposes a 197.3 MW wind facility interconnecting at the South Woodrow 100 kV substation in the Michigan area. The request was received on April 25, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 186 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$95,509,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Boyles – Unc Qf 100.0 kV	J R Whiting 230-1 kV XFMR	300.9	107	52.1
Boyles Substation – Black Hills Power 100.0 kV	J R Whiting 230 kV/1 kV Transformer Bank	280.4	121	52.1
Carrollton-Upfield – Unc Qf 100.0 kV	J R Whiting 230/1kV Transformer Bank	273.5	115	52.1
South Woodrow – Black Hills Power 100.0 kV	J R Whiting 230/1kV Auto	271.7	130	52.1
Koch Oil – Deer Park 100.0 kV	J R Hiting 230/1kV Autotransformer	251.2	122	52.1
Carrollton-Upfield Substation – Koch Oil 100.0 kV	J R Whiting 230-1 kV XFMR	224.3	133	52.1
Deer Park – Corbell 100.0 kV	J R Whiting 230-1 kV XFMR	186.8	152	52.1
Broken Bow – Corbll 100.0 kV	J R Whiting 230-1 kV XFMR	164.7	172	52.1
Pickord – Herald-Twin Cities 230.0 kV	Trego – Beaver 500.0 kV	149.1	664	11.9
Gray'S Flat Substation – Baraa Vlg 230.0 kV	Madeline – East Methuen 230.0 kV line	146.4	519	8.0
Larson St – Norwich 230.0 kV	Barrett – Norwich Substation 500.0 kV	145.0	668	8.9
Onancock Diesel Generation Facility – Joso 230.0 kV	Beaver – Eufaula 500.0 kV	144.0	795	17.3
Highgate – Herald-Twin Cities 230.0 kV L	Mason St 500/230kV TX	143.8	518	9.2
Adams – Pickford 230.0 kV Line	Trego – Beaver 500.0 kV L	140.0	531	9.8
Canalrd – Weedatch 230.0 kV	Trego – Beaver 500.0 kV line	138.7	513	8.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Newport Tie – Post Rock 230.0 kV	Trego Substation – Beaver 500.0 kV	136.7	864	14.2
Lewisville Switch – Doyle Energy Facility 100.0 kV	J R Whiting 230-1 kV TX	134.2	143	26.9
Onancock Diesel Generation Facility – Adams 230.0 kV	Trego – Beaver 500.0 kV L	131.9	1,112	20.0
Post Rock – Myrtle Point 230.0 kV	Trego – Beaver 500.0 kV Line	131.4	931	13.6
J R Whiting 230 kV/1 kV TX	County Line 19 – South Woodrow 100.0 kV	130.9	387	69.3
Madeline – E Methuen 230.0 kV line	Baraga Vlg – Gray'S Flat 230.0 kV	129.2	607	8.5
J R Whiting 100-1 kV XFMR	S Woodrow – County Line 19 100.0 kV	129.0	393	69.3
Onancock Diesel Generation Facility – Beaver 230.0 kV	Beaver – Eufaula 500.0 kV	125.5	911	11.5
E Methuen 500/230kV XFMR	Swan Vly – Trego 500.0 kV	123.6	981	9.4
New Departure – Adams 230.0 kV	Trego – Beaver 500.0 kV L	122.6	613	10.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Boyles – Unc Qf 100.0 kV	Rebuild line with higher-capacity conductor	\$3,870,000	78% (\$3,038,000)
Black Hills Power – Boyles 100.0 kV L	Rebuild line with higher-capacity conductor	\$7,574,000	78% (\$5,946,000)
Unc Qf – Carrollton-Upfield 100.0 kV	Rebuild line with higher-capacity conductor	\$6,214,000	78% (\$4,879,000)
Black Hills Power – S Woodrow 100.0 kV L	Rebuild line with higher-capacity conductor	\$4,899,000	78% (\$3,846,000)
Koch Oil – Deer Park 100.0 kV	Rebuild line with higher-capacity conductor	\$2,068,000	78% (\$1,624,000)
Carrollton-Upfield – Koch Oil 100.0 kV	Rebuild line with higher-capacity conductor	\$1,689,000	78% (\$1,326,000)
Deer Park – Corbell 100.0 kV	Rebuild line with higher-capacity conductor	\$527,000	78% (\$414,000)
Broken Bow – Corbell 100.0 kV	Rebuild line with higher-capacity conductor	\$2,471,000	78% (\$1,940,000)
Pickford – Herald-Twin Cities 230.0 kV	Rebuild line with higher-capacity conductor	\$58,031,000	18% (\$10,429,000)
Baraga Vlg – Gray'S Flat 230.0 kV L	Rebuild line with higher-capacity conductor	\$18,450,000	11% (\$1,952,000)
Larson St – Norwich 230.0 kV	Rebuild line with higher-capacity conductor	\$9,346,000	11% (\$1,056,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Onancock Diesel Generation Facility – Joso 230.0 kV	Rebuild line with higher-capacity conductor	\$22,886,000	16% (\$3,604,000)
Highgate – Herald-Twin Cities 230.0 kV	Rebuild line with higher-capacity conductor	\$22,943,000	27% (\$6,186,000)
Adams – Pickford 230.0 kV Line	Rebuild line with higher-capacity conductor	\$21,532,000	17% (\$3,630,000)
Canalrd – Weedatch 230.0 kV	Rebuild line with higher-capacity conductor	\$21,054,000	11% (\$2,234,000)
Newport Tie – Post Rock 230.0 kV	Rebuild line with higher-capacity conductor	\$39,574,000	11% (\$4,475,000)
Lewisville Switch – Doyleenergy Facility 100.0 kV	Rebuild line with higher-capacity conductor	\$2,890,000	62% (\$1,790,000)
Adams Substation – Onancock Diesel Generation Facility 230.0 kV	Rebuild line with higher-capacity conductor	\$33,774,000	17% (\$5,665,000)
Myrtle Pt – Post Rock 230.0 kV Line	Rebuild line with higher-capacity conductor	\$26,457,000	12% (\$3,107,000)
J R Whiting 230/1kV Transformer	Replace transformer with higher-rated unit	\$9,458,000	100% (\$9,458,000)
East Methuen – Madeline Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$30,182,000	11% (\$3,216,000)
J R Whiting 100-1 kV XFMR	Replace transformer with higher-rated unit	\$3,894,000	100% (\$3,894,000)
Onancock Diesel Generation Facility – Beaver 230.0 kV	Rebuild line with higher-capacity conductor	\$23,240,000	13% (\$3,107,000)
E Methuen 500/230kV XFMR	Replace transformer with higher-rated unit	\$24,561,000	7% (\$1,816,000)
Adams Substation – New Departure Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$32,707,000	21% (\$6,877,000)
	Total allocated to project		\$95,509,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.